$$\sigma_x \otimes \tau_z = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}.$$

Notice that $\sigma_x \otimes \tau_z$ is not the same as $\sigma_z \otimes \tau_x$. That is natural, because they represent different observables.

So far, so good. But next, we'll see something a little more interesting. With the help of a few exercises, we'll try to convince you that the Kronecker product really *is* the tensor product for matrices—in other words, that Alice's half of the matrix only affects her half of the column vector, and likewise for Bob. This is tricky because of the way the Kronecker product mixes up the elements of its building blocks.

As an example, let's look at how $\sigma_z \otimes \tau_x$ acts on $|ud\rangle$. Translating the abstract symbols into components, we can write

$$(\sigma_z \otimes \tau_x)|ud\rangle = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

But the column vector on the right-hand side corresponds to $|uu\rangle$ in Eqs. 7.9. Translated back into abstract notation, this becomes

$$(\sigma_z \otimes \tau_x)|ud\rangle = |uu\rangle.$$

This is exactly what we want—a matrix representation of our abstract operators and state-vectors that replicates their known behavior.

The following exercise will help crystallize the idea that the σ-half of $\sigma\otimes\tau$ only affects Alice's half of the state-vector, and that the τ-half only affects Bob's. The one after that provides some practice working out the matrix